Azure Cloud Fundamentals and Data Pipeline Implementation Using Azure Data Factory (ADF)

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Objective

The objective of this assignment is to understand Microsoft Azure cloud services and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory. The assignment includes creating Azure resources, storing data in Blob Storage, building a data pipeline, validating metadata, assigning IAM roles, and monitoring pipeline execution.

Tools and Services Used

Service	Purpose
Microsoft Azure Portal	Cloud resource management
Azure Resource Group	Logical grouping of resources
Azure Storage Account	Cloud storage service
Azure Blob Storage	Store CSV files
Azure Data Factory	Build and execute data pipelines
Azure IAM	Access management and permissions

Task 1 – Explore Azure Portal

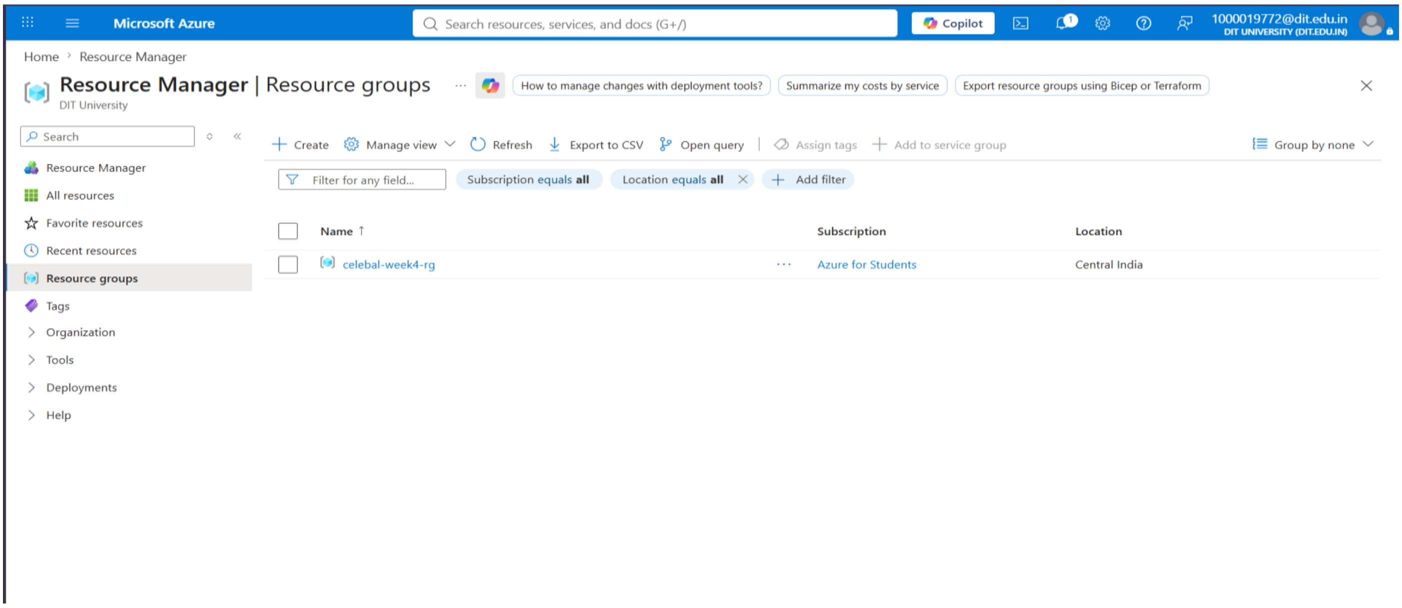
Objective

Create a Resource Group to organize all Azure resources used during the assignment.

Procedure

1. Logged into the Microsoft Azure Portal.
2. Opened the **Resource Groups** service.
3. Created a new Resource Group named **celebal-week4-rg**.
4. Selected **Central India** as the deployment region.
5. Verified successful creation of the Resource Group.

Screenshot:



Task 2 – Storage Setup

Objective

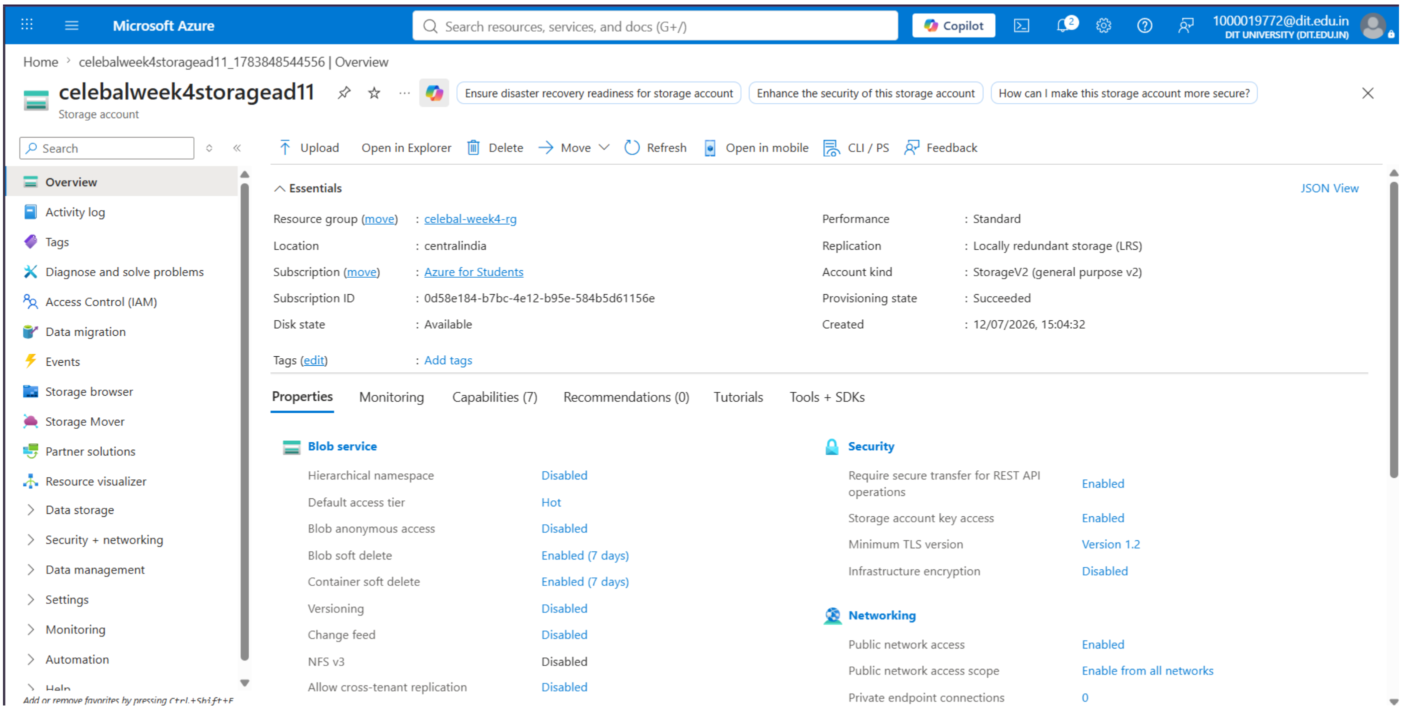
Create an Azure Storage Account, configure Blob Storage containers, and upload the source CSV dataset.

Procedure

Step 1: Create Storage Account

1. Opened Azure Storage Accounts.
2. Created a storage account named: **celebalweek4storagead11**
3. Selected:
   - Resource Group: **celebal-week4-rg**
   - Region: **Central India**
   - Performance: Standard
   - Replication: Locally Redundant Storage (LRS)

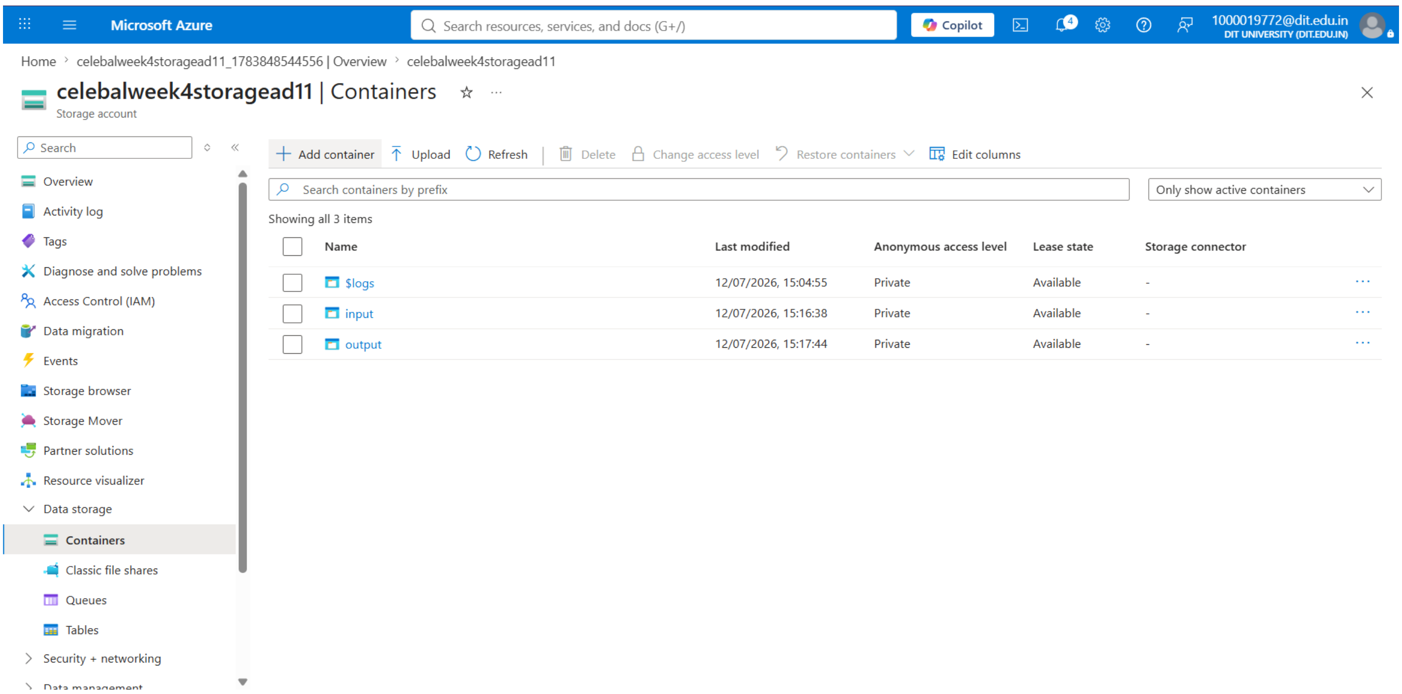
Screenshot:



Step 2: Create Blob Containers

Three Blob Containers were created: $logs , input, output

Screenshot:



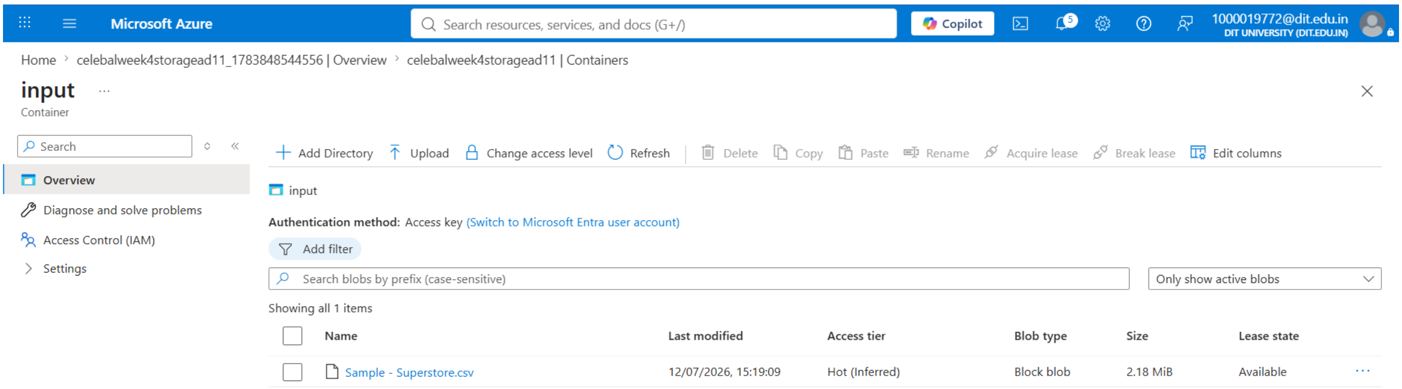
Step 3: Upload Source Dataset

The CSV file:

Sample - Superstore.csv was uploaded into the **input** container.

The uploaded dataset serves as the source file for subsequent Azure Data Factory operations.

Screenshot:



Task 3 – Azure Data Factory Basics

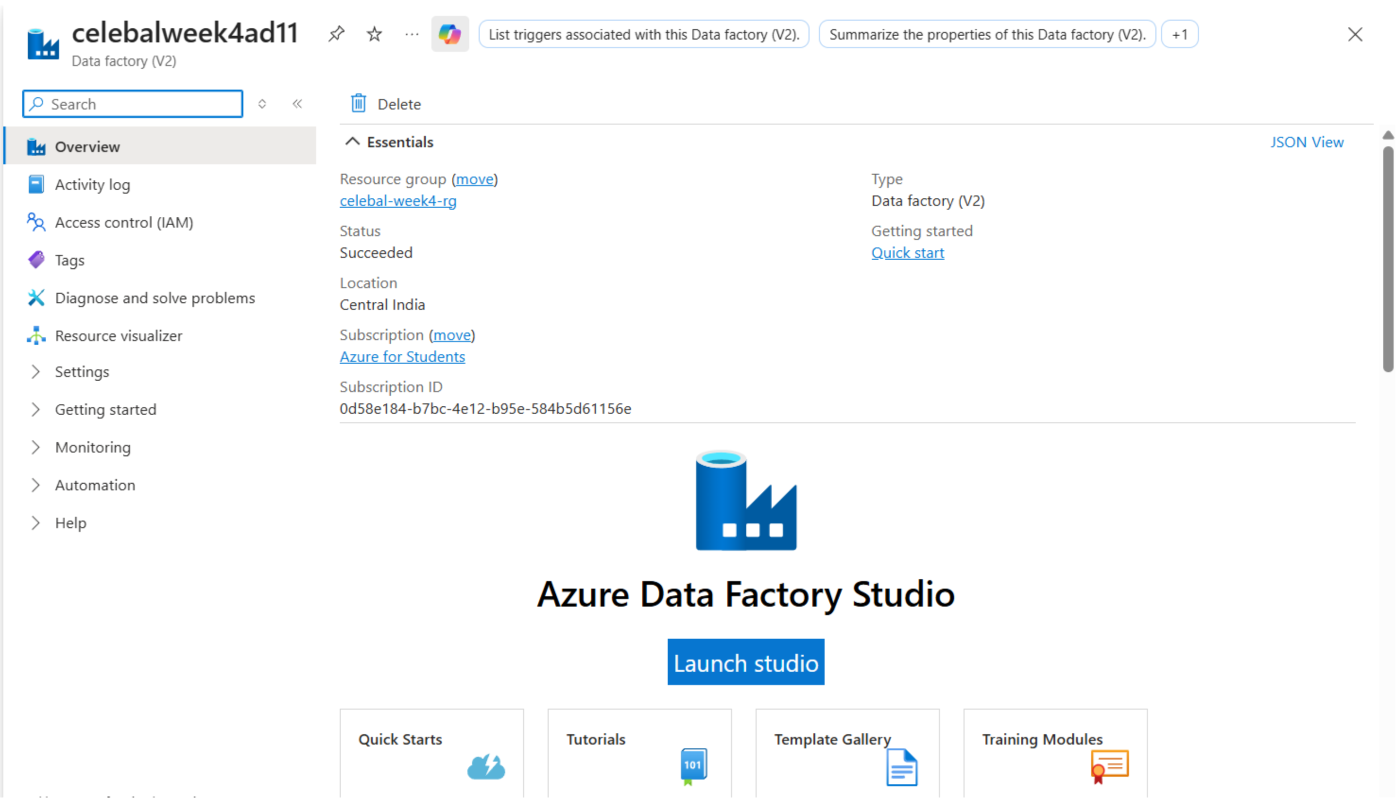
Objective

Create Azure Data Factory and configure all components required for building data pipelines.

Step 1: Create Azure Data Factory

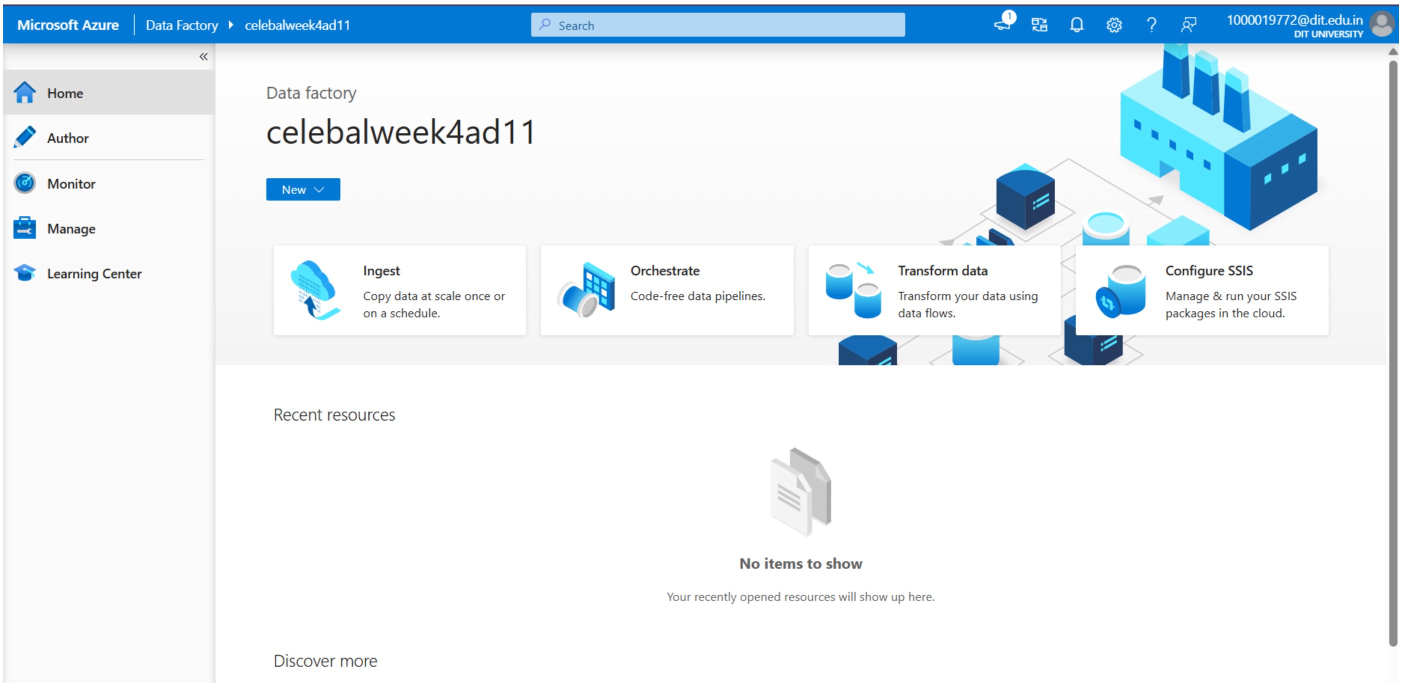
A new Azure Data Factory instance named **celebalweek4ad11** was created inside the existing Resource Group.

Screenshot:



After deployment, Azure Data Factory Studio was launched for authoring and monitoring pipelines.

Screenshot:

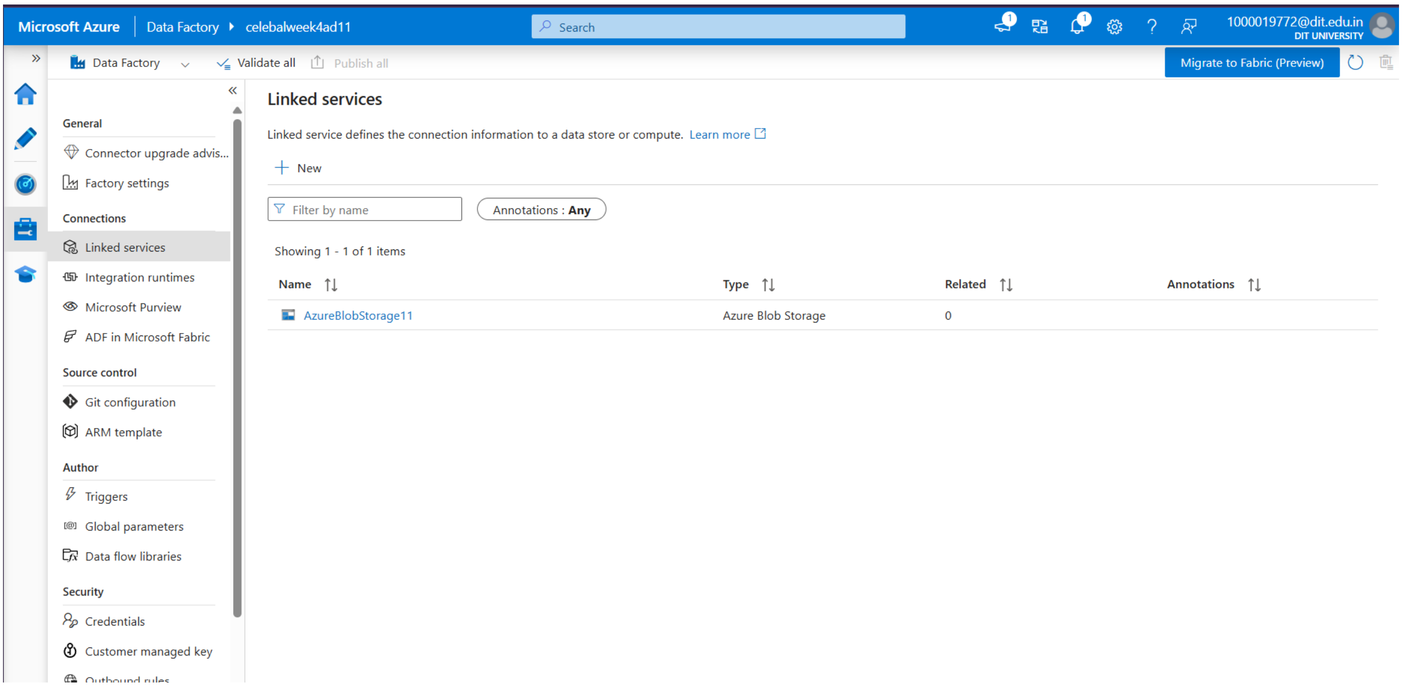


Step 2: Create Linked Service

A Linked Service establishes the connection between Azure Data Factory and Azure Blob Storage.

A linked service named **AzureBlobStorage11** was created using Azure Blob Storage authentication.

Screenshot:



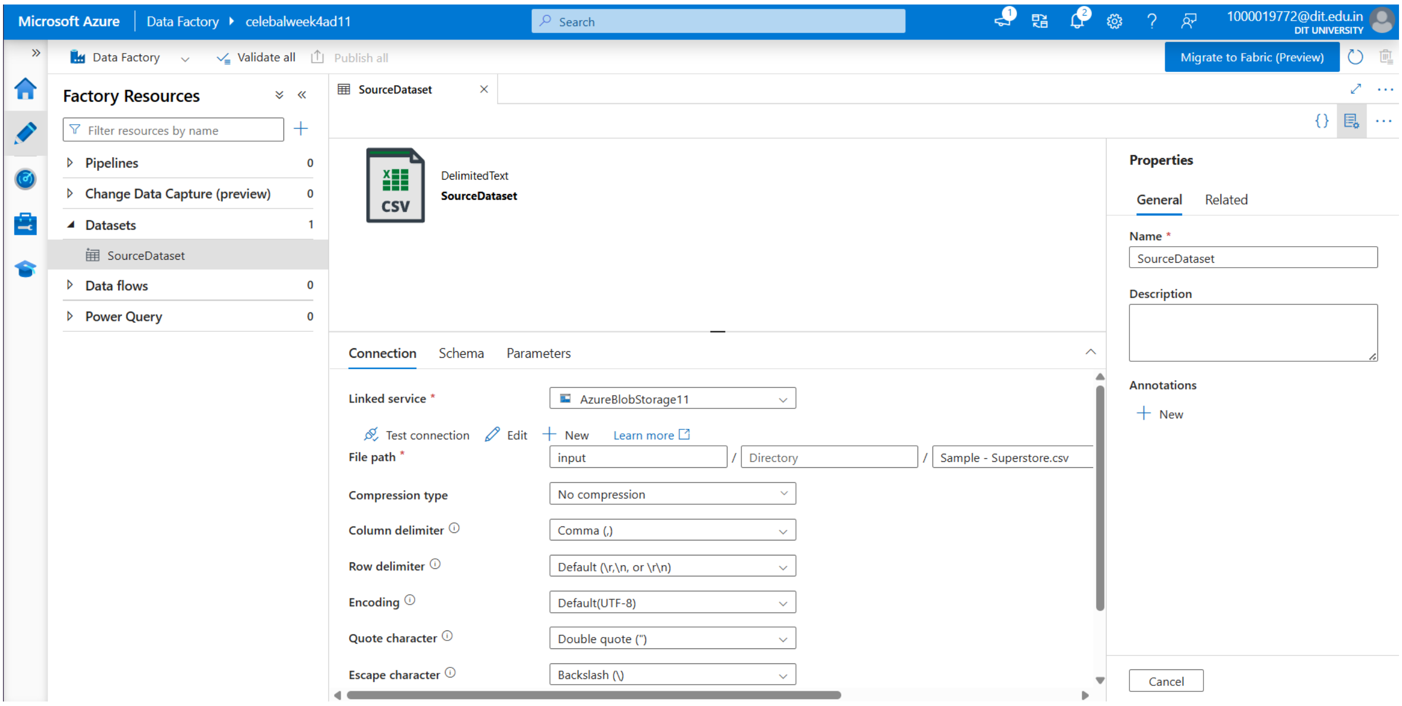
Step 3: Create Source Dataset

A dataset named **SourceDataset** was created with the following configuration:

- Linked Service: AzureBlobStorage11
- Container: input
- File: Sample - Superstore.csv
- File Type: Delimited Text (CSV)

This dataset represents the source data used by the pipelines.

Screenshot:

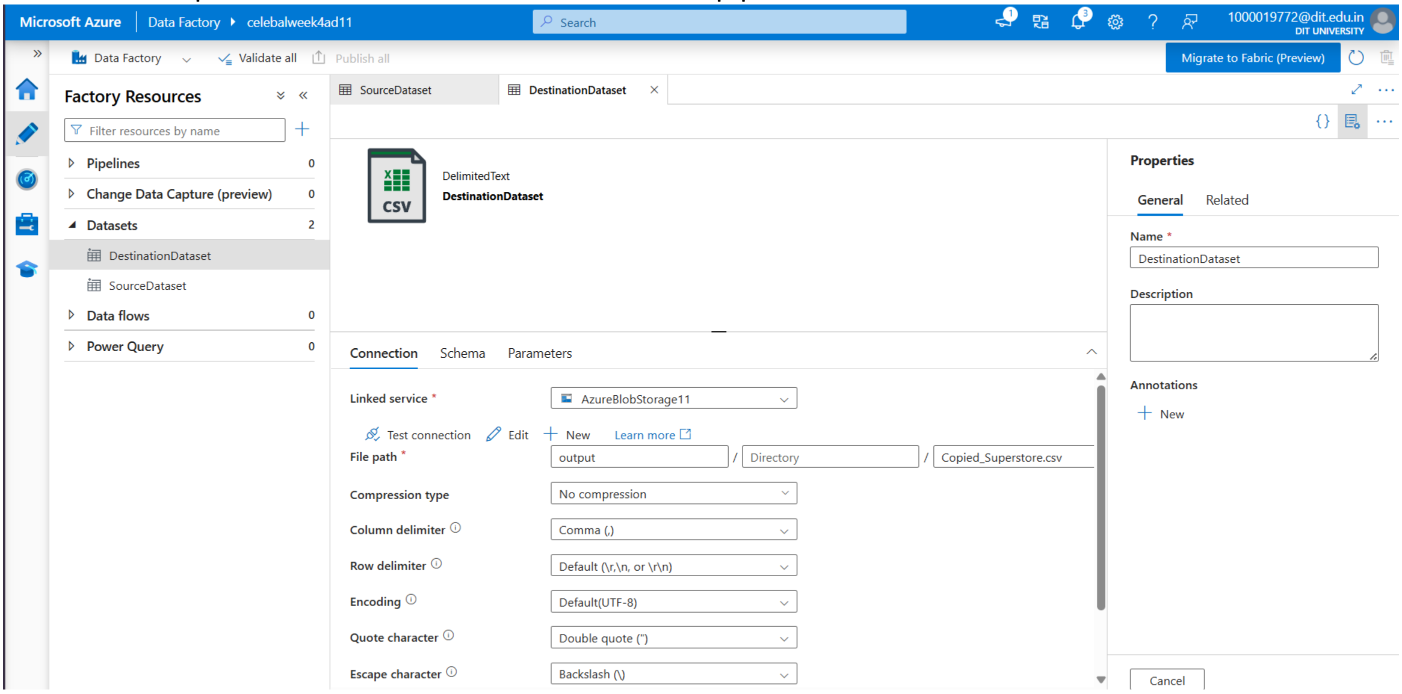


Step 4: Create Destination Dataset

A second dataset named **DestinationDataset** was created for storing copied data.
Configuration:

- Linked Service: AzureBlobStorage11
- Container: output
- File: Copied_Superstore.csv

This dataset represents the destination location of the pipeline.

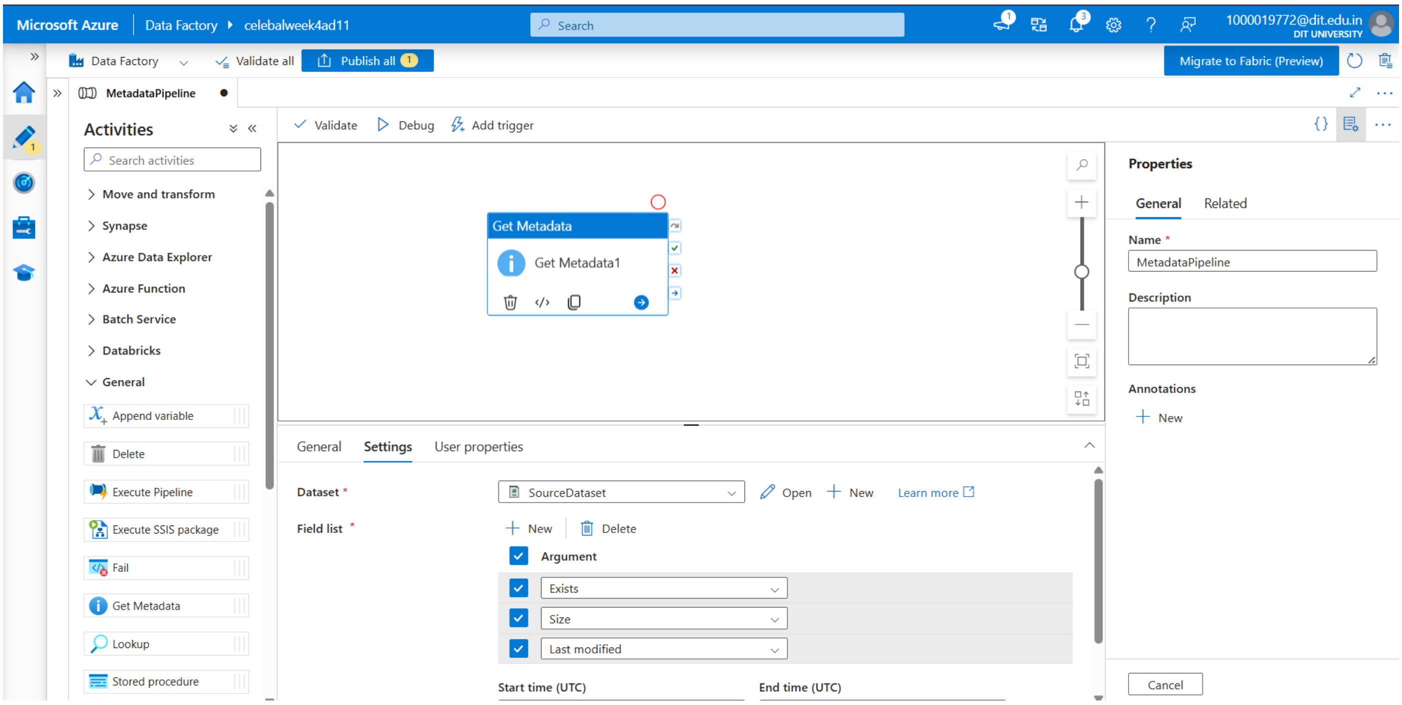


Step 5: Create Metadata Pipeline

A pipeline named **MetadataPipeline** was created using the **Get Metadata** activity.
The SourceDataset was selected, and the following metadata properties were configured:

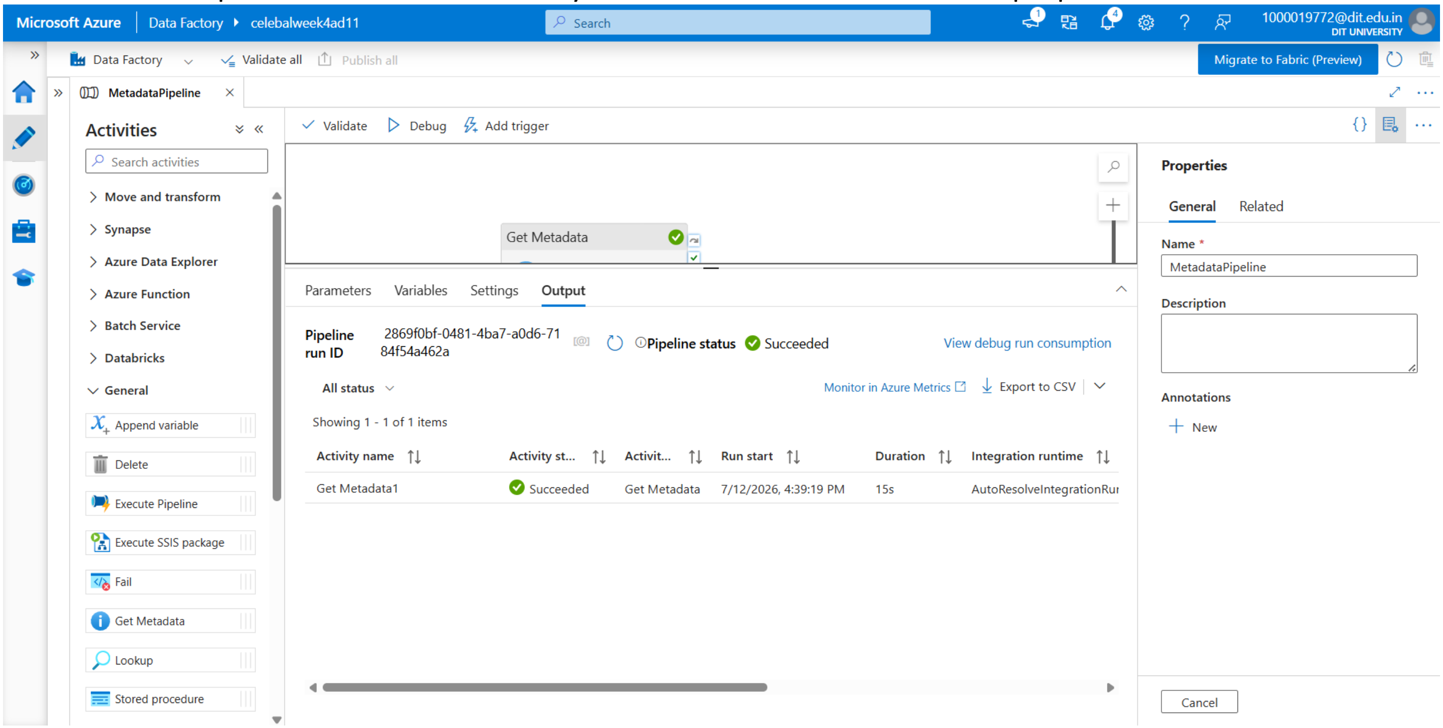
- Exists
- Size
- Last Modified

Screenshot:



Result:

The Metadata Pipeline executed successfully and validated the existence and properties of the source dataset.



Task 4 – Pipeline Development

Objective

Develop a pipeline to copy the source CSV file from one Blob Storage container to another.

Procedure

A pipeline named **CopyPipeLine** was created using the **Copy Data** activity.

Configuration:

Source:

- Dataset: SourceDataset

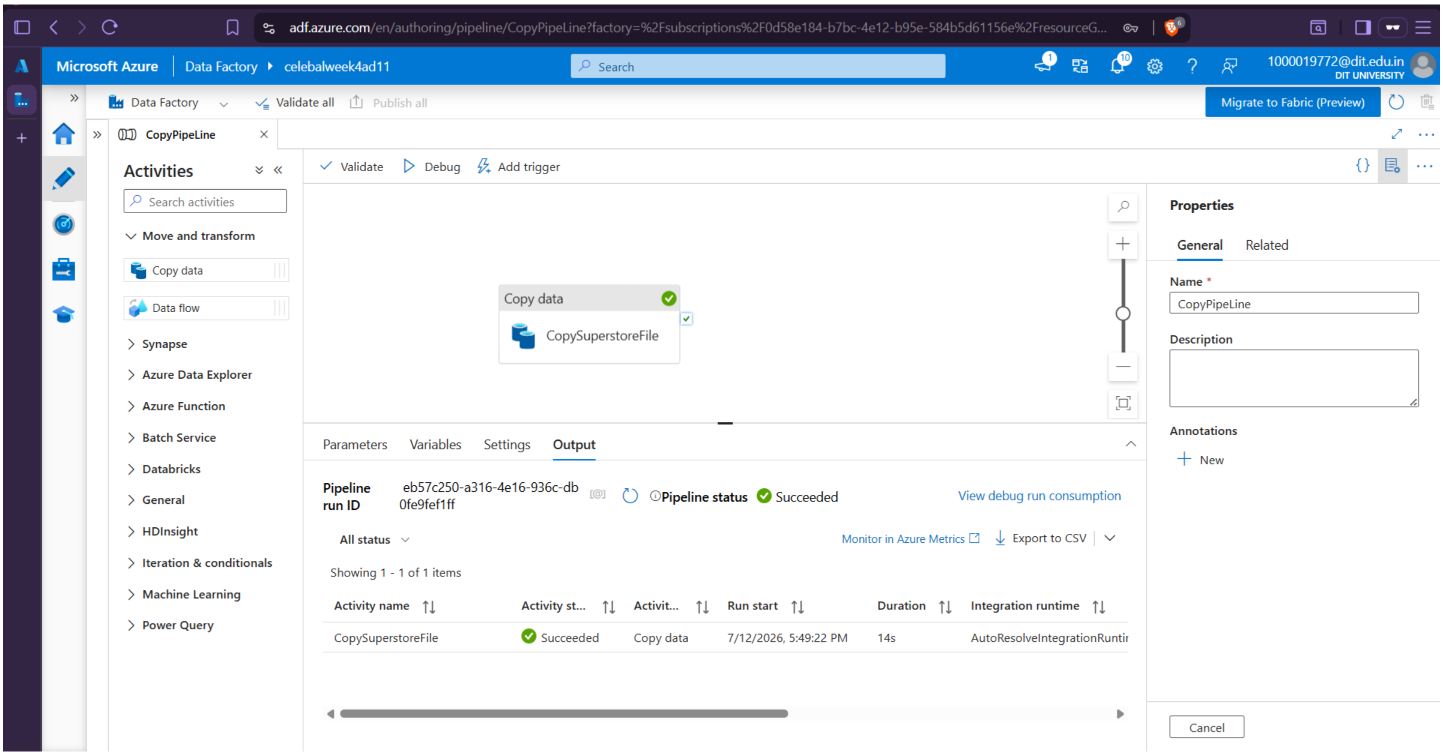
Destination:

- Dataset: DestinationDataset

The Copy Data activity transfers the CSV file from the **input** container to the **output** container.

The pipeline was validated successfully before execution.

Screenshot:



Task 5 – Pipeline Execution

Objective

Execute the developed pipeline and verify successful completion.

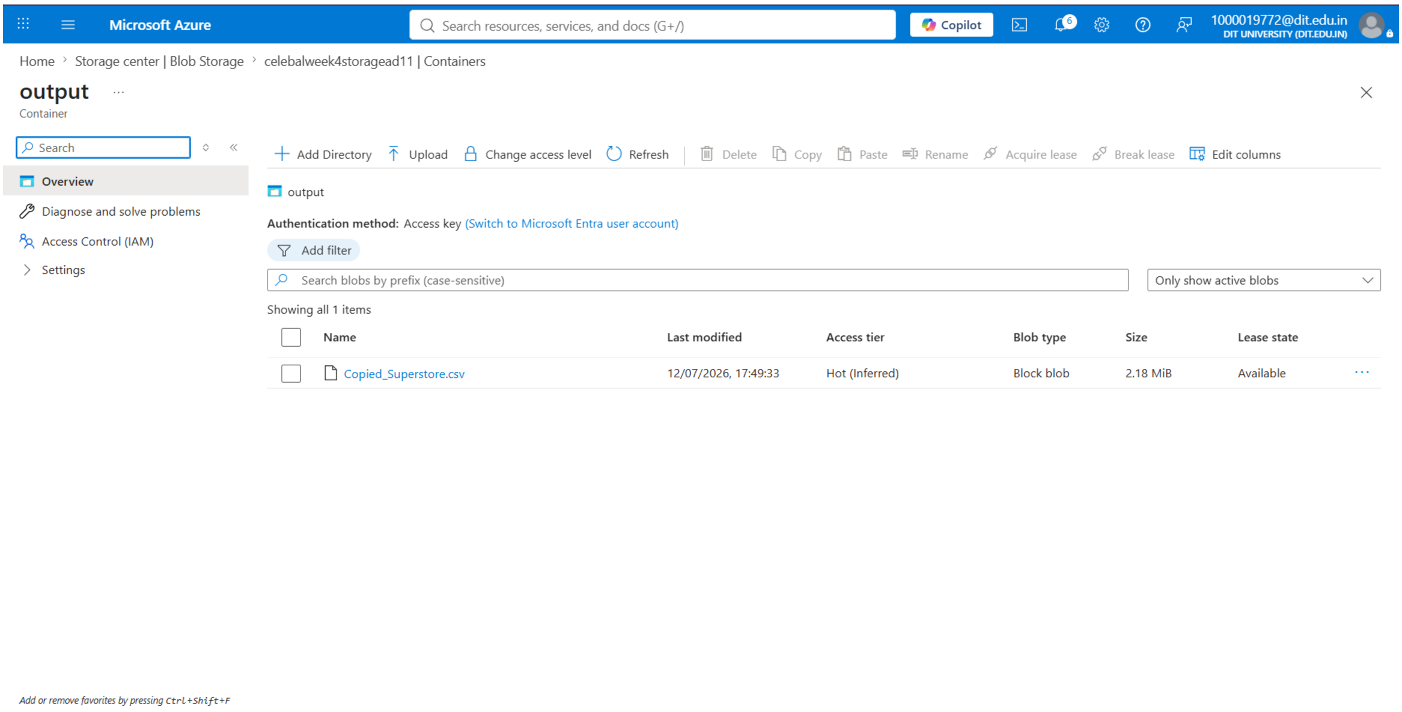
Procedure

1. Opened CopyPipeLine.
2. Clicked **Debug** to execute the pipeline.
3. Monitored execution status.
4. Confirmed successful completion.
5. Verified that the destination file was generated inside the output container.

Result

The pipeline completed successfully.
The file **Copied_Superstore.csv** was created in the destination Blob Storage container.

Screenshot:



Task 6 – IAM Roles

Objective

Configure Azure Identity and Access Management permissions.

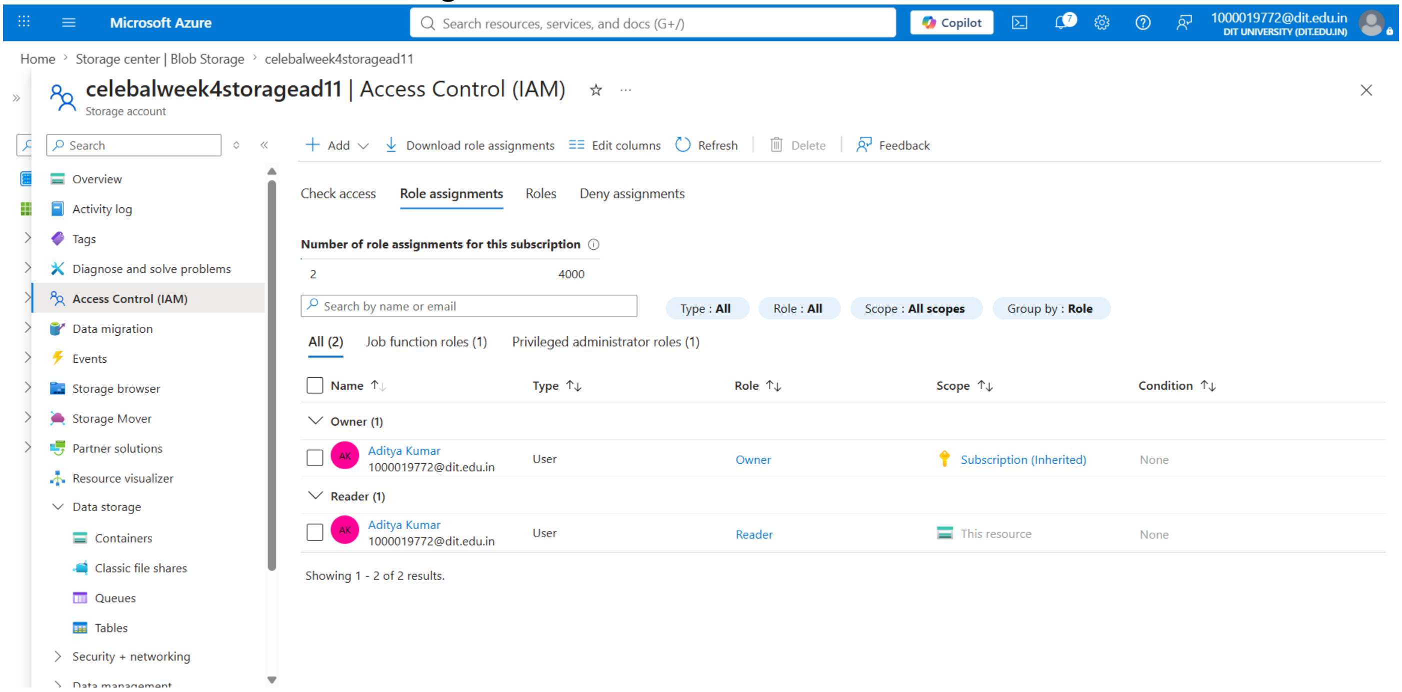
Procedure

The following roles were configured:

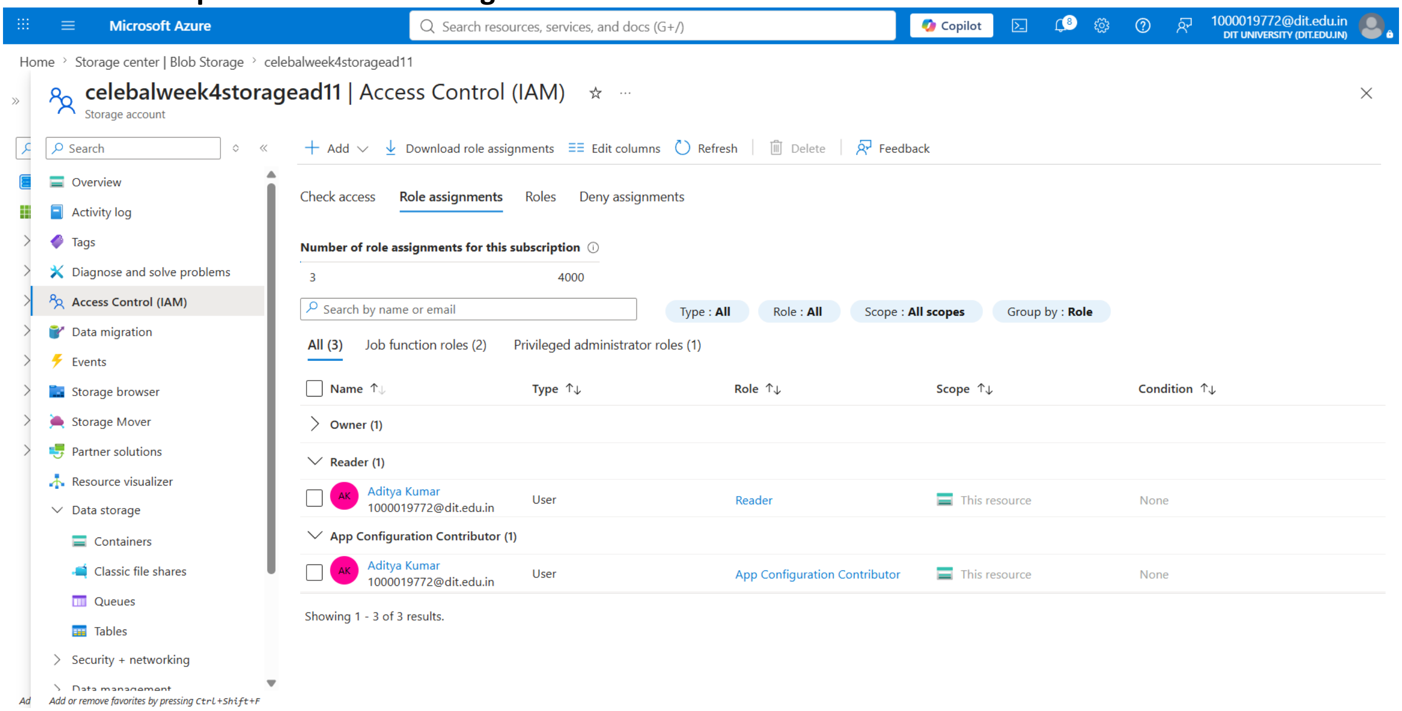
- Owner
- Reader
- App Configuration Contributor

These permissions ensured appropriate access to the Storage Account and Azure Data Factory resources throughout the implementation.

Screenshot - Initial IAM Role Assignments:



Screenshot - Updated IAM Role Assignments:



Mini Project

Problem Statement

Develop a complete Azure Data Factory solution capable of:

- Reading a CSV file from Azure Blob Storage
- Validating file metadata
- Copying the file to another Blob Storage location
- Successfully executing the workflow

Implementation

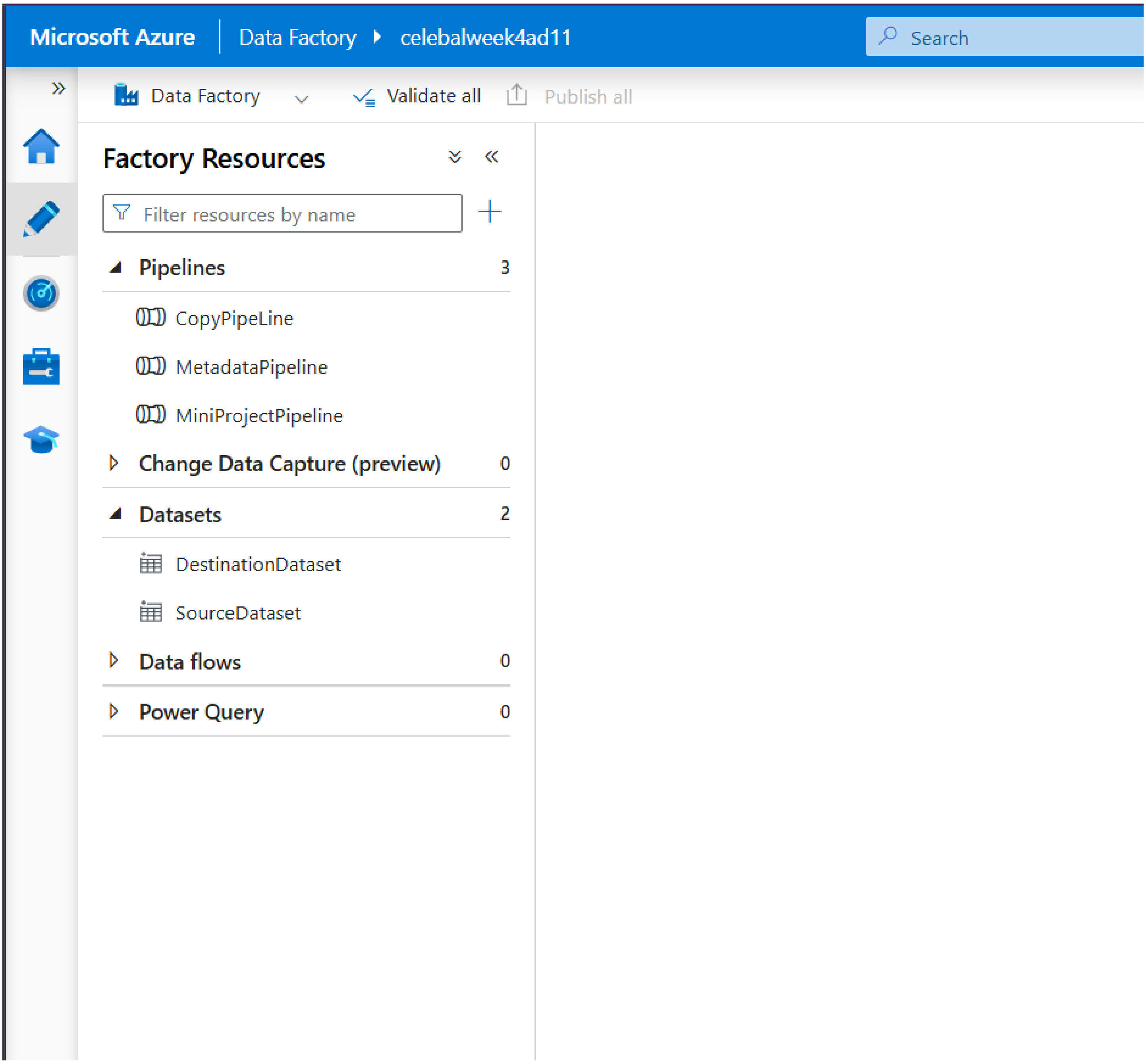
A pipeline named **MiniProjectPipeline** was developed to combine metadata validation and file copying into a single automated workflow.

Pipeline sequence:

1. Validate source file using **Get Metadata**
2. Upon successful validation, execute **Copy Data**
3. Store copied file in destination Blob Storage

The workflow follows a success dependency, ensuring that file copying occurs only after metadata validation completes successfully.

Screenshot:

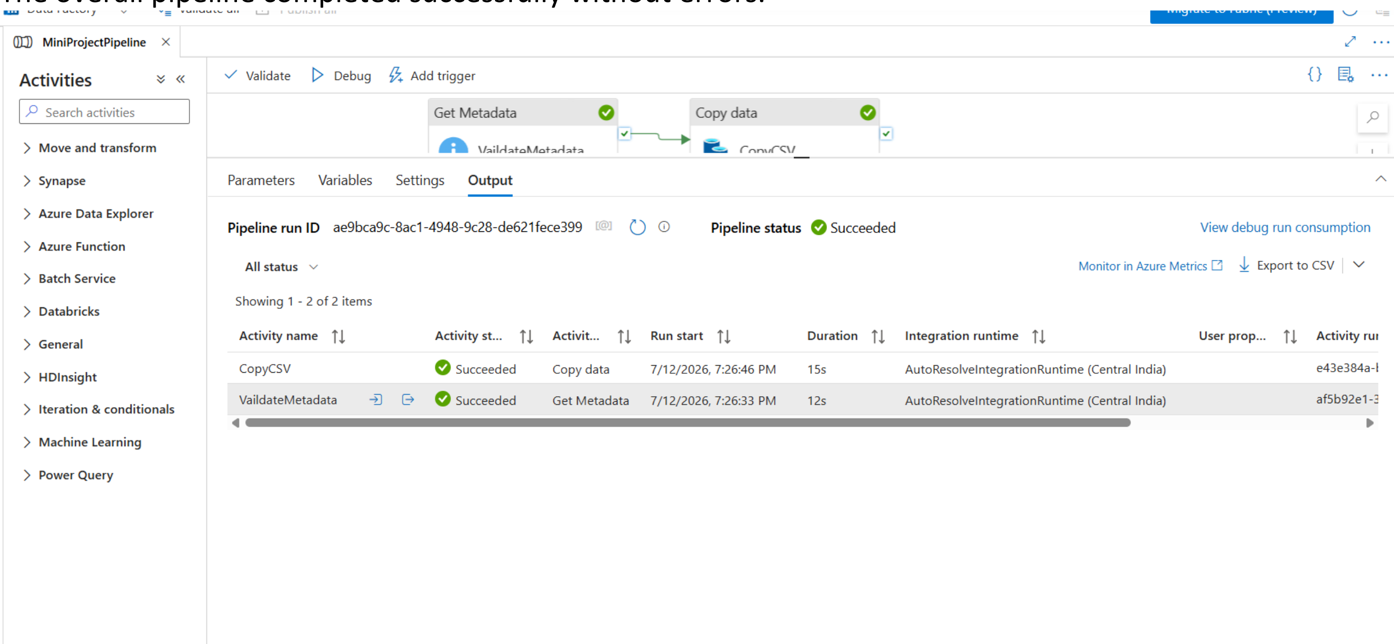


Pipeline Execution

The MiniProjectPipeline was executed using Debug mode.
Execution Results:

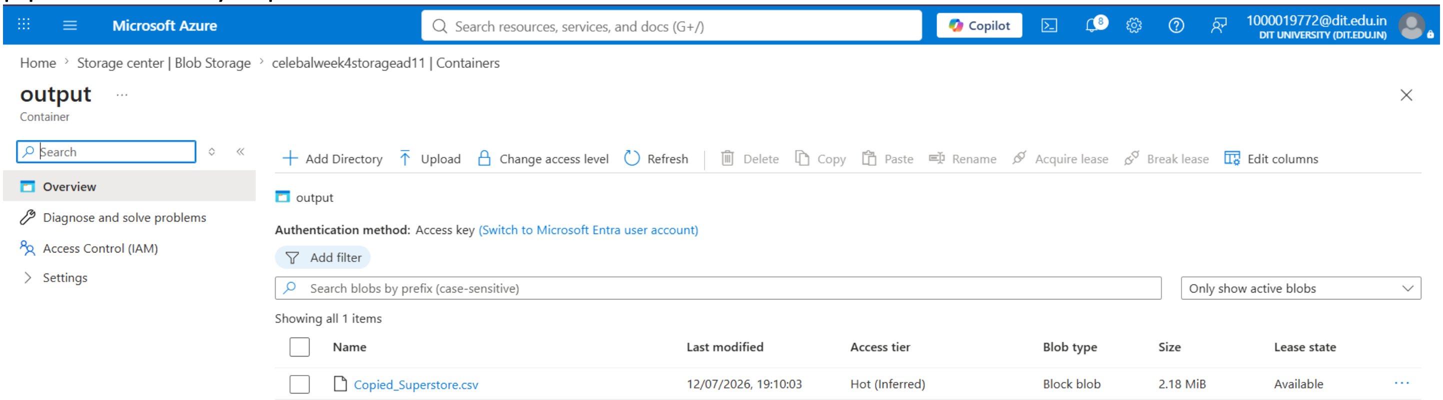
- Metadata Validation – Succeeded
- Copy Data – Succeeded

The overall pipeline completed successfully without errors.



Output Verification

After successful execution, the output container contained **Copied_Superstore.csv** confirming that the pipeline correctly copied the source dataset.



Conclusion

This assignment provided practical exposure to Microsoft's Azure cloud platform and Azure Data Factory. Beginning with the creation of cloud resources, Blob Storage was configured to store source and destination datasets, while Azure Data Factory was used to orchestrate metadata validation and file copying through reusable pipelines. Appropriate IAM permissions were configured to ensure secure access to cloud resources.

The final mini project successfully integrated metadata validation and data copying into a single automated workflow, demonstrating the fundamental concepts of cloud storage, data integration, workflow orchestration, and pipeline execution. The successful completion of all tasks provided hands-on experience with Azure services commonly used in real-world cloud data engineering and ETL solutions.